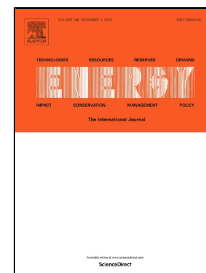


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Gas turbine performance prediction via machine learning

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Abstract

This paper develops a machine learning-based method to predict gas turbine performance for power generation. Two surrogate models based on high dimensional model representation (HDMR) and artificial neural network (ANN) are developed from real operational data to predict the operating characteristics of air compressor and turbine. Both models capture the operating characteristics well with average errors of less than 1.0%. Moreover, four more holistic models are developed to capture gas turbine part-load and full-load performance. The models for air compressor and turbine are then embedded into a gas turbine simulation program, and all surrogate models are validated using a separate data set. It is shown that the power output, pressure ratio, fuel flow, and turbine exhaust temperature from these models match their measured values well with average and maximum errors of less than 2.0% and 4.3%, respectively. Since holistic ANN models have lower complexity and higher accuracy, the ANN model for predicting full-load performance is used to construct gas turbine performance correction curves. The correction curves along with the ANN model for predicting part-load performance offer an excellent basis for continuous health monitoring and fault diagnosis. The proposed methodology is applicable to any gas turbines and can help power plants to study and quantify performance degradation over time.

Keywords: Gas turbine, Surrogate models, Machine learning, Performance prediction, Correction curves, Simulation

1 Introduction

Gas turbines (GTs) have been widely used for power generation due to their high thermal efficiency and low CO₂ emissions. GT installations are steadily rising to meet the soaring electricity demand resulted from the growing populations and economies [1]. GTs normally run in a flexible load-following manner due to the increasing penetration of renewable energy (e.g. solar energy, wind energy, and biomass energy). This operation mode requires GTs to ramp up and down frequently, and thus a precise prediction of GT performance is crucial for maintaining efficient operation. To this end, accurate and reliable GT modeling methods are necessary.

GT modeling has been studied in the literature. Al-Hamdan and Ebaid [2] studied the modeling and simulation of a GT for power generation by superimposing a turbine map on a compressor map via suitable axis transformations. Haglind and Elmegaard [3] proposed two models for predicting the part-load performance of aero-derivative GTs. One used actual maps for the compressor and turbine, while the other utilized a compressor map with turbine constants. Both models were found to offer good predictions of mass flow and pressure ratio profiles over 40-100% loads as well as thermal efficiencies and exhaust temperatures for part-loads above 70%. Lee et al. [4] developed a general simulation program for simple, recuperative, and reheat GTs. A stage-stacking method with a general stage map was used for the compressor, while a stage-by-stage model with efficiency correction was adopted for the turbine. However, their program was validated only under full-load conditions, and thus its prediction accuracy under part-load conditions was questionable. Song et al. [5] established a performance analysis model for a three-spool GT using detailed compressor and turbine maps and showed that the model offered good accuracy in predicting GT performance. Although these models can predict GT performance, they all require detailed performance maps of air

compressor and turbine [6]. In practice, real performance maps are proprietary information of engine manufacturers and often unknown even to the GT users. Therefore, the models discussed so far are unable to provide reliable and accurate prediction, and there is a need to develop models from real operational data for predicting GT performance.

Artificial neural network (ANN) is a powerful surrogate model and has been applied to predict GT performance. Bartolini et al. [7] utilized ANN to describe micro GT performance, including completing the performance diagrams, assessing the influences of ambient parameters, and predicting the pollutant emissions. Nikpey et al. [8] presented an ANN model for a micro GT and showed that it was able to predict its normal performance with high accuracy. Later, they used ANN to evaluate the performance and emissions of a recuperative micro GT burning mixtures of natural gas and biogas [9]. Bhowmik et al. [10] developed an ANN model for a diesel engine fueled with ternary blends of diesel, kerosene, and ethanol. Their model was found to be capable of mapping the ternary blends with engine performance and exhaust emissions. Fast et al. [11] unified two ANN models to predict the performance of an industrial GT. Their models were trained to identify if anti-icing system was in operation, and then predict GT operational and performance parameters. Their results showed good prediction accuracy with an error of less than 2.0%. Gonca [12] utilized ANN to analyze the effects of GT design parameters on the energetic, exergetic, and ecological performance of a GT with two intercoolers and two re-heaters. Moreover, ANN was applied to other GT-based applications [13–17].

The above ANN models can predict GT performance; however, they are unable to detect if GTs run at full-load conditions and thus cannot predict GT full-load performance. Knowing GT full-load performance, especially power generation capacity, is useful, because it helps GT users determine the operational limits of their engines. With GT power generation

capacity, GT users can respond to the dynamic power demands flexibly and claim more reserve capacity to the power grid to gain more profit, if the power grid mandates such reserves. Hence, it is desirable for GT users to accurately predict GT full-load performance.

Generally, compressor and turbine performance maps are needed to predict GT performance. However, these performance maps are proprietary information of engine manufacturers and even unknown to GT users. A promising remedy is to use real operational data to predict the compressor and turbine operating characteristics (not their full performance maps) or to directly map GT performance with proper operating parameters. Machine learning methods based on surrogate models can be employed to achieve these goals. The resulting models can then be used to predict GT part-load and full-load performances. Typically, GT full-load performance can be expressed in terms of correction curves that represent a departure of GT performance from its reference performance. Correction curves allow GT users to determine GT performance without appealing to complex numerical simulations and have been used for on-line performance monitoring [18] and diagnosis [19]. Considering GT correction curves are proprietary and may often be unknown to GT users, deriving them from GT operational data via reverse engineering can be very useful.

In this paper, a supervised machine learning-based method is developed for predicting GT performance and constructing performance correction curves. Surrogate models based on high dimensional model representation (HDMR) and artificial neural network (ANN) are developed to obtain the operating characteristics of air compressor and turbine from their real operational data. Moreover, holistic HDMR and ANN surrogate models are developed to capture GT part-load and full-load performance. The HDMR and ANN models for air compressor and turbine are then embedded into a GT simulation program for performance prediction. All surrogate models are validated using real GT operational data. Finally, a

holistic ANN model is employed to construct GT performance correction curves due to its lower complexity and higher accuracy.

2 Methodology

Our proposed method for predicting GT performance is developed and demonstrated for a heavy-duty GT in a power plant. However, it is applicable to any other GTs. The GT comprises a multi-stage axial-flow air compressor, a dry low- NO_x combustion chamber, and a multi-stage expansion turbine. The ambient air passes through an air filter to the compressor. The pressurized air from the compressor enters the combustion chamber, where natural gas as a fuel is burnt with the air. The resulting hot gas expands in the turbine to produce power. Some air is extracted from the proper stages of the compressor and then precooled for turbine blade cooling. However, since detailed cooling parameters are unavailable, turbine blade cooling is approximated by bleeding some air from the compressor discharge and then precooled by ambient air to cool the turbine rotor blades in this study, as shown in Fig. 1.

2.1 Data collection and screening

A distributed control system (DCS) adjusts the inlet guide vane (IGV) opening (θ) and fuel flow (m_f) to regulate GT operation and meet the power demand. Along with θ and m_f , DCS records compressor inlet and outlet pressures (P_1, P_2) and temperatures (T_1, T_2), fuel pressure (P_f), fuel temperature (T_f), lower heating value (LHV_f), turbine exhaust pressure (P_4), turbine exhaust temperature (TET), GT shaft speed (N), and GT power (W_{GT}). In this paper, N is constant at 3000 rpm, the pressure loss in the combustion chamber is assumed to be 5% of its inlet pressure, and the turbine always works under choking conditions.

The measured DCS data are screened to remove the transients during any start-up and shut-down operations. Moreover, all corrupt data with obvious sensor signal errors are removed. Hence, 2675 clean data points reflecting the GT steady-state operation are obtained.

Since DCS does not record mass flow of air (m_1) and turbine inlet temperature (T_3), they are estimated from the measured data using a GT simulation program built in GateCycle [20]. The turbine cooling air flow (m_{ca}) is determined by the following empirical formula (Eq. (1)) from the GT industry.

$$m_{ca}(\text{kg s}^{-1}) = 0.057533m_1 - 0.378318 \quad (1)$$

2.2 Input and output parameters

The compressor and turbine operating characteristics are described by their performance maps as shown in Fig. 3, where corrected mass flow and corrected speed are defined as follows [21–23]:

$$\text{Corrected mass flow: } m_{cor} = m_{in} \sqrt{T_{in}} / P_{in} \quad (2)$$

$$\text{Corrected speed: } N_{cor} = N / \sqrt{T_{in}} \quad (3)$$

where, m is the mass flow, T is the temperature, P is the pressure, subscript cor denotes the corrected value, and in denotes the inlet. The compressor map normally expresses the corrected mass flow ($m_{c,cor}$) and isentropic efficiency (η_c) as functions of pressure ratio (PR) and corrected speed ($N_{c,cor}$) with IGV fully open ($\theta = 100\%$) as follows.

$$m_{c,cor} = h_1(PR, N_{c,cor}) \text{ and } \eta_c = h_2(PR, N_{c,cor}) \quad (4)$$

Similarly, the turbine map expresses $m_{t,cor}$ and η_t as functions of expansion ratio (π) and corrected speed ($N_{t,cor}$).

$$m_{t,cor} = h_3(\pi, N_{t,cor}) \text{ and } \eta_t = h_4(\pi, N_{t,cor}) \quad (5)$$

Since DCS adjusts θ and m_f to regulate GT power, compressor operation is directly affected by θ and m_f . This means both $m_{c,cor}$ and η_c depend on θ , m_f and $N_{c,cor}$. Since N is constant and η_c determines T_2 , we propose the following functional relations.

$$m_{c,cor} = f_1(\theta, T_1, m_f), \text{ and } T_2 = f_2(\theta, T_1, m_f) \quad (6)$$

For the turbine, π and $N_{t,cor}$ will determine turbine operation, namely $m_{t,cor}$ and η_t . Since N is constant, T_3 determines $N_{t,cor}$, and η_t determines TET . This leads to the following from Eq. (5).

$$m_{t,cor} = f_3(\pi, T_3) \text{ and } TET = f_4(\pi, T_3) \quad (7)$$

f_1 to f_4 are unknown since compressor and turbine performance maps are unavailable; however, they can be approximated by surrogate models using real operational data. In this work, HDMR and ANN are used as the surrogate models except for f_3 . Since the turbine always works under choking condition [see the horizontal line in Fig. 2(b)], $f_3(\pi, T_3) = m_{t,cor}$ is constant, and its value is the average of all calculated corrected mass flows as shown in Fig. 3. Moreover, compressor pressure ratio (PR) is determined via matching compressor and turbine operation under the choking condition [21,22].

2.3 HDMR models

HDMR method [24] is a powerful surrogate modeling technique, since it considers potential non-linearity and contributions from the interactions of input variables. It expands a full function [$y = f(x)$] as a sum of functions, each of which depends on subsets of M input variables ($x_1, x_2, \dots, x_M$) as follows:

$$y = f_0 + \sum_{i=1}^M f_i(x_i) + \sum_{i=1}^M \sum_{j=i+1}^M f_{ij}(x_i, x_j) + \dots + f_{123\dots M}(x_1, x_2, \dots, x_M) \quad (8)$$

where, i and j index the input variables, and f_0 is the mean value of $f(x)$. The expansion in Eq. (8) has a finite number of terms and represents $f(x)$ exactly. However, for practical engineering applications, terms with more than two input variables can often be ignored due to their negligible contributions [25,26]. Hence, Eq. (9) is often sufficient for constructing an HDMR surrogate model.

$$y \approx f_0 + \sum_{i=1}^M f_i(x_i) + \sum_{i=1}^M \sum_{j=i+1}^M f_{ij}(x_i, x_j) \quad (9)$$

Numerical integration can be used to evaluate each of these terms; however, a more efficient alternative is to approximate $f_i(x_i)$ and $f_{ij}(x_i, x_j)$ in Eq. (9) using orthogonal polynomials [27,28], as shown in Eq. (10). Numerical software like MATLAB [29] can be used to obtain the parameters (C , $A_{i,k}$, and $B_{i,j,k,l}$) in Eq. (10) from the operational data.

$$y = C + \sum_{i=1}^M \sum_{k=1}^K A_{i,k} \times x_i^k + \sum_{i=1}^M \sum_{j=i+1}^M \sum_{k=1}^K \sum_{l=1}^K B_{i,j,k,l} \times x_i^k \times x_j^l \quad (10)$$

In brief, three HDMR models are developed in this work to predict the operating characteristics of the compressor and turbine with input and output parameters shown in Table 1 as follows.

HDMR-1(a-b): Predict $m_{t,cor}$ and T_2 respectively from θ , T_1 , and m_f . (Compressor)

HDMR-1c: Predict T_4 from π and T_3 . (Turbine)

Moreover, holistic HDMR models are developed to predict GT part-load and full-load performance. For any ambient temperature T_1 , GT operation is fully determined if W_{GT} is given, as DCS adjusts θ and m_f to produce W_{GT} as per its own operating strategy. Hence, for predicting GT part-load performance, HDMR-2 models use T_1 and W_{GT} as input parameters and PR , m_f , T_4 , and m_4 as output parameters. The full-load performance of a GT occurs when IGV is fully open (i.e. $\theta = 100\%$) and turbine inlet temperature (T_3) is maintained at its maximum value via adjusting m_f . Therefore, HDMR-3 models employ θ , T_1 , and m_f as input parameters and W_{GT} , PR , T_3 , T_4 , and m_4 as output parameters for predicting GT full-load performance.

In brief, nine HDMR models are developed to predict GT part-load and full-load performance as follows.

HDMR-2(a-d): Predict PR , m_f , T_4 , and m_4 respectively from T_1 and W_{GT} . (Reverse prediction)

HDMR-3(a-e): Predict W_{GT} , PR , T_3 , T_4 , and m_4 respectively from θ , T_1 , and m_f . (Forward prediction)

Using Eq. (10) with the 2675 measured data, the three HDMR models for predicting the operating characteristics of the compressor and turbine are obtained in Table 2. The coefficients of determination (R^2) exceed 0.98, which suggest excellent surrogate qualities. The prediction accuracy in Table 3 is fairly good for all output responses as more than 98.0% (2623 out of 2675) of the data points have errors less than 2.0%. Moreover, their average errors are all less than 1.0%. The holistic HDMR models (HDMR-2 and HDMR-3) for predicting GT part-load and full-load performances are presented in Table 4 and Table 6, respectively. The surrogate qualities are fairly good as indicated by their coefficients of determination. The prediction accuracies in Table 5 and Table 7 suggest that the holistic HDMR models predict GT performance very well, as more than 94.7% (2535 out of 2675) of the data points have errors less than 2.0% and the average errors for all output responses are all less than 1.0%. All these suggest that the HDMR models can capture the complex operations of the compressor, turbine, and GT.

2.4 ANN models

ANN is an adaptive computation system that mimics the biological network of the human brain. It learns the relationships between inputs and outputs through an iterative training process. ANN comprises a number of artificial neurons with linear and nonlinear activation functions and thus is well capable of predicting the nonlinear behavior of a system. In recent years, multi-layer perceptron (MLP) feedforward neural networks consisting of an input layer, one or more hidden layer(s), and an output layer are widely used for modeling, monitoring, and diagnosis of GTs [7,11] and power plants [30,31].

MLP with one hidden layer, as shown in Fig. 4, has been proved to be able to approximate any continuous nonlinear function, as long as a sufficient number of hidden neurons is used [32]. Hence, in this study, MLP with one hidden layer is employed to obtain the operating characteristics of the compressor and turbine. The resulting ANN models are referred as ANN-1 in this paper. Table 1 shows the input and output parameters for ANN-1 models. Moreover, holistic ANN models (ANN-2 and ANN-3) are developed to predict GT part-load and full-load performance as HDMR-2 and HDMR-3 do. Their input and output parameters are the same as HDMR-2 and HDMR-3.

In brief, four ANN models are developed in this work with the following input and output parameters.

ANN-1a: Predict $m_{t,cor}$ and T_2 from θ , T_1 , and m_f . (Compressor)

ANN-1b: Predict T_4 from π and T_3 . (Turbine)

ANN-2: Predict PR , m_f , T_4 , and m_4 from T_1 and W_{GT} . (Reverse prediction)

ANN-3: Predict W_{GT} , PR , T_3 , T_4 , and m_4 from θ , T_1 , and m_f . (Forward prediction)

All ANN models are developed in MATLAB. *tansig* and *purelin* activation functions are used for the hidden and output layers respectively, because of their successful application in GT performance prediction [8,33]. The input and output data are normalized to $[-1, +1]$ for adapting to *tansig* function. The ANN models are trained using a gradient descent with momentum and adaptive learning rate backpropagation (*traingdx*) algorithm with mean squared error (MSE) as the loss function. The data are randomly divided into two subsets: 80% for training and 20% for validation. ANN is trained with 2, 4, 6, ..., 18, and 20 hidden neurons to find the best network structure. For each hidden neuron number, network weights and biases are optimized five times to ensure a global optimum. The detailed ANN training parameters are given in Table 8. Using the 2675 measured data, the best networks for ANN-

1a, ANN-1b, ANN-2, and ANN-3 are determined to be (3-6-2), (2-6-1), (2-6-4), and (3-8-5), respectively.

Tables 9-11 show the prediction errors of the ANN models for the compressor, turbine, and GT. Fairly accurate predictions are achieved, since more than 94.4% (2526 out of 2675) the data points have errors less than 2.0%. Furthermore, their average errors are all less than 1.0%. This indicates that the ANN models can represent the operating characteristics of the compressor, turbine, and GT quite well.

2.5 Evaluation and comparison of surrogate models

The reliability of the surrogate models is evaluated using an independent set of 217 data points. HDMR-1 and ANN-1 for the compressor and turbine are embedded into a GT simulation program built in GateCycle [20] for performance prediction. Under part-load conditions, GT operation is fixed, if W_{GT} is given and T_1 is known, as DCS adjusts θ and m_f as per its own operating strategy to produce W_{GT} . Hence, θ and m_f are manipulated in the GateCycle program augmented with HDMR-1 and ANN-1 to produce W_{GT} . To this end, a simulation-based optimization method [34] is employed to minimize m_f as shown in Fig. 5. GateCycle simulates the GT and MATLAB does the optimization. Nonlinear Optimization by Mesh Adaptive Direct search (NOMAD) is used as the optimizer. GateCycle and MATLAB are interfaced via ActiveX for direct two-way communication. Moreover, HDMR-2 and ANN-2 directly take T_1 and W_{GT} for performance evaluation (reverse evaluation). For predicting GT full-load performance, θ , T_1 , and m_f can be directly used in HDMR-3, ANN-3, and the augmented GateCycle programs to evaluate their performance (forward evaluation). For both reverse and forward evaluations, PR , m_f , T_4 , and W_{GT} computed from the surrogate models are compared with their measured values to obtain the relative errors as shown in Fig. 6 and Fig. 7, respectively.

Fig. 6 presents the relative errors of PR , m_f , and TET for HDMR-1 (HDMR-2, ANN-1, ANN-2) from reverse evaluation. The average errors of the three operating parameters are 0.91% (0.72%, 1.43%, 0.72%), 1.20% (0.47%, 1.57%, 0.54%), and 0.58% (0.40%, 0.72%, 0.38%), respectively. Moreover, their maximum errors are 3.35% (3.06%, 3.38%, 3.10%), 2.84% (2.14%, 3.15%, 2.15%), and 1.46% (1.20%, 1.56%, 1.01%), respectively. The average errors of W_{GT} , PR , and TET for HDMR-1 (HDMR-3, ANN-1, ANN-3) from forward evaluation are 1.48% (0.50%, 1.93%, 0.61%), 0.44% (0.33%, 0.46%, 0.35%), and 0.44% (0.34%, 0.59%, 0.35%), respectively, as shown in Fig. 7. The corresponding maximum errors are 3.02% (2.02%, 4.23%, 2.32%), 1.67% (1.23%, 1.76%, 1.45%), and 1.18% (1.07, 1.28%, 1.00%), respectively. From Fig. 6 and Fig. 7, we can conclude that all surrogate models predict GT performance quite well. However, the holistic models are slightly more accurate than HDMR-1 and ANN-1.

$$AIC = n \ln \left(\frac{SSE}{n} \right) + 2p \quad (11)$$

where, AIC is the Akaike information criterion, n is the number of data points, SSE is the sum of squared error, and p is the number of model parameters.

Akaike information criterion (AIC) is applied to assess the relative quality and complexity of the surrogate models [35,36]. Table 12 presents a comparison of HDMR and ANN models. As can be seen, ANN-2 and ANN-3 have the lowest AIC values, followed by HDMR-2 and HDMR3 as well as HDMR-1 and ANN-1, respectively. This suggests that ANN-2 and ANN-3 are relatively better than the other surrogate models. Since ANN-2 and ANN-3 are also easy to use, we recommend ANN-2 for predicting part-load performance while ANN-3 for predicting full-load performance.

3 GT correction curves

GT performance varies with ambient and operating conditions (e.g. ambient temperature). For GT users, it is desirable to know both part-load and full-load performance of their engines at any time. Full-load performance occurs when IGV remains fully open ($\theta = 100\%$) and T_3 is maintained at its maximum value via adjusting m_f . GT users will know the maximum T_3 of their engines, otherwise it can be estimated from the GT operating data. For this case study, 1266 °C is taken as the maximum T_3 from GT simulation. In this work, the actual GT power varied from 125 MW to 220 MW at various operating conditions. Fig. 8 shows the actual power and maximum possible power that could have been generated from the GT at each time (data point) as predicted by the surrogate models. As can be seen, the GT is mostly running at part-load conditions. ANN-2 can be easily used to predict this part-load performance. Moreover, the GateCycle program augmented with HDMR-1 and ANN-1 predicts a relatively lower power generation capacity, while HDMR-3 and ANN-3 give similar predictions. This is mainly because HDMR-1 and ANN-1 introduce internal errors, when they are embedded into GateCycle program for performance prediction. Hence, HDMR-3 and ANN-3 are better for predicting the GT power generation capacity. However, since ANN-2 and ANN-3 are relatively better as shown in Table 12, we use them for predicting GT performance. Fig. 9 presents the normalized GT part-load performance from ANN-2. As can be seen, the variation trends of various performance indicators agree well with those in [23]. Moreover, ANN-3 is utilized to reproduce GT correction curves for predicting full-load performance as discussed in the following.

GT correction curves represent the variation in GT performance with the operating conditions (e.g. ambient temperature). They can be used to predict full-load performance at any operating condition. However, these curves are proprietary information of engine

manufacturers and thus may be unavailable to the GT users. Hence, it is useful to reproduce the GT correction curves. For this, a reference condition is needed for correcting GT performance. In this study, ANN-3 is used to reproduce these curves with GT full-load performance at $T_1 = 30^\circ\text{C}$ as the reference operating point, as shown in Table 13.

Fig. 10 presents the GT correction curves with respect to ambient temperature (T_1). GT power, efficiency, and turbine exhaust flow (TEF) can be predicted using the correction curves as follows [37,38].

$$CF_X = CF_{X,P} \times CF_{X,T} \times CF_{X,RH} \quad (12)$$

$$W_{GT} = W_{GT,ref} \times CF_W \quad (13)$$

$$\eta_{GT} = \eta_{GT,ref} \times CF_\eta \quad (14)$$

$$TEF = TEF_{ref} \times CF_{TEF} \quad (15)$$

where, CF is the correction factor, subscript ref denotes the reference condition, and X denotes GT power, efficiency, or turbine exhaust flow. The overall correction factor is the product of all individual correction factors (e.g. ambient pressure, ambient temperature, and relative humidity). Eqs. (16-17) are used to predict the turbine exhaust temperature (TET). In this work, $CF_{X,P}$ and $CF_{X,RH}$ are taken as unity, while $CF_{TET,P}$ and $CF_{TET,RH}$ are taken as null, since ambient pressure and relative humidity are not measured during GT operation.

$$CF_{TET} = CF_{TET,P} + CF_{TET,T} + CF_{TET,RH} \quad (16)$$

$$TET = TET_{ref} + CF_{TET} \quad (17)$$

Using ANN-2 along with the GT correction curves, GT users can predict the part-load and full-load performance of their engines at any given operating condition. Hence, they can assess the operational status and health condition of their engines without appealing to complex and demanding numerical computations or simulations. Moreover, they can apply these surrogate models to identify potential operational problems (e.g. compressor fouling)

and thus maintain good equipment health. More importantly, knowing the current operational status and power generation capacity of their engines at any time, the GT users can respond flexibly and quickly to the dynamic power demand and thus maintain efficient GT operation. Furthermore, the proposed method can be applied to historical annual data to derive performance degradation curves for GTs.

4 Conclusions

This paper proposed a method utilizing HDMR and ANN surrogate models to achieve supervised machine learning based on historical data to predict GT performance. HDMR and ANN models were developed to predict the operating characteristics of the compressor and turbine and then embedded into a GT simulation program for performance prediction. Moreover, holistic HDMR and ANN models were developed to capture GT part-load and full-load performance. All surrogate models were found to capture GT overall part-load and full-load performance quite well on confidential operational data. Holistic ANN models showed relatively better quality than the other surrogate models. Furthermore, HDMR-3 and ANN-3 gave the best predictions for the maximum power generation capacity. Since ANN-3 had relatively better quality, it was used to reproduce GT correction curves, which along with ANN-2 can predict GT part-load and full-load performance under varying operating conditions. These models can offer a basis for continuous health monitoring and operational problem diagnosis. Finally, the developed method can be applied to yearly operational data to study and quantify performance degradation of GTs.

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Nomenclature

Symbols

CF	correction factor
LHV	lower heating value, kJ/kg
m	mass flow, kg/s
n	number of data points
N	shaft speed, rpm
p	number of model parameters
P	pressure, bar
PR	pressure ratio
T	temperature, K
W	power, kW
θ	IGV opening
η	efficiency
π	expansion ratio
SSE	sum of squared error
TEF	turbine exhaust flow

TET turbine exhaust temperature

Subscripts

a air

c compressor

ca cooling air

cor corrected value

ref reference condition

f fuel

t turbine

Acronyms

AIC Akaike information criterion

ANN artificial neural network

DCS distributed control system

GT gas turbine

HDMR high dimensional model representation

IGV inlet guide vane

MLP multi-layer perceptron

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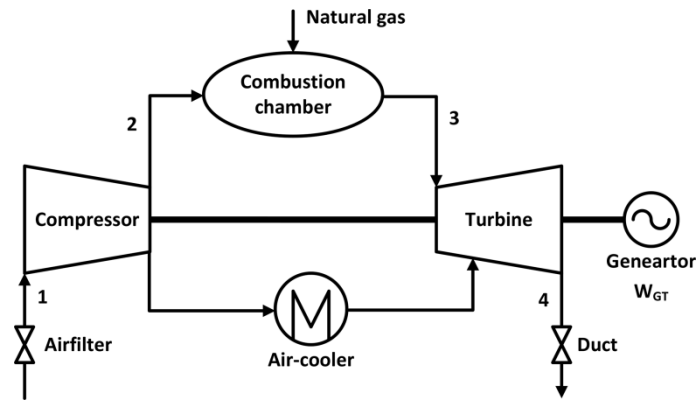


Fig. 1 Schematic of a heavy-duty gas turbine.

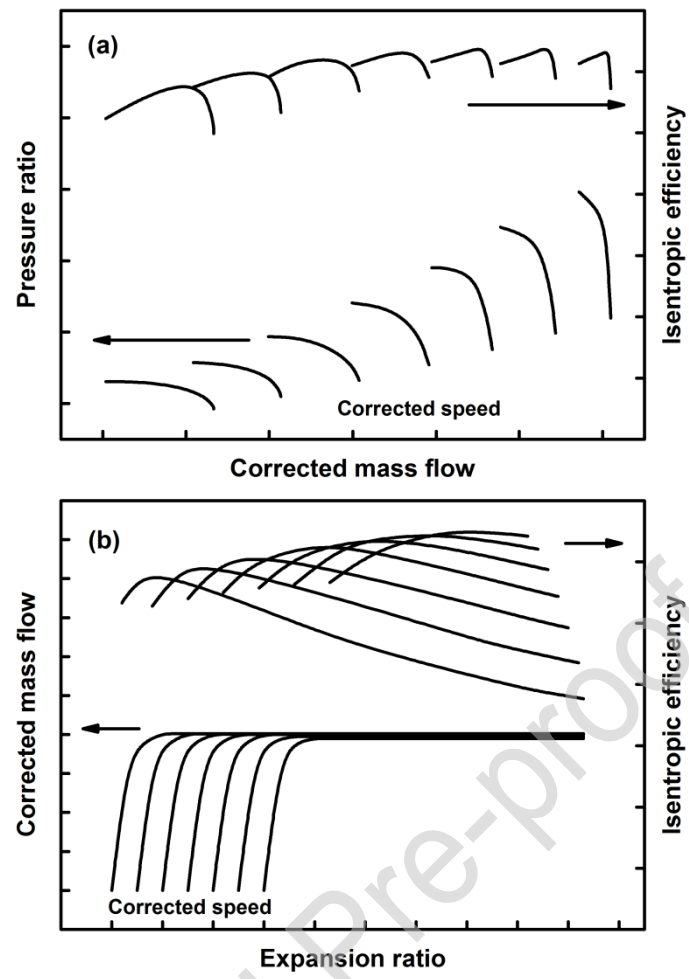


Fig. 2 Compressor map (a) and turbine map (b).

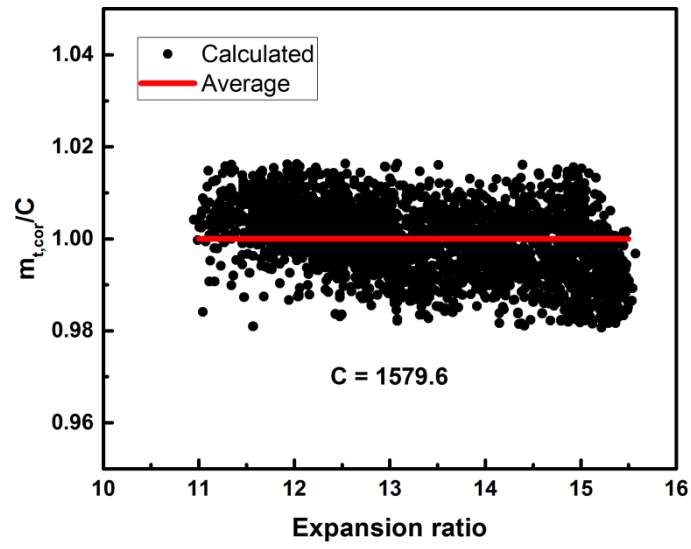


Fig. 3 Turbine corrected mass flow versus expansion ratio.

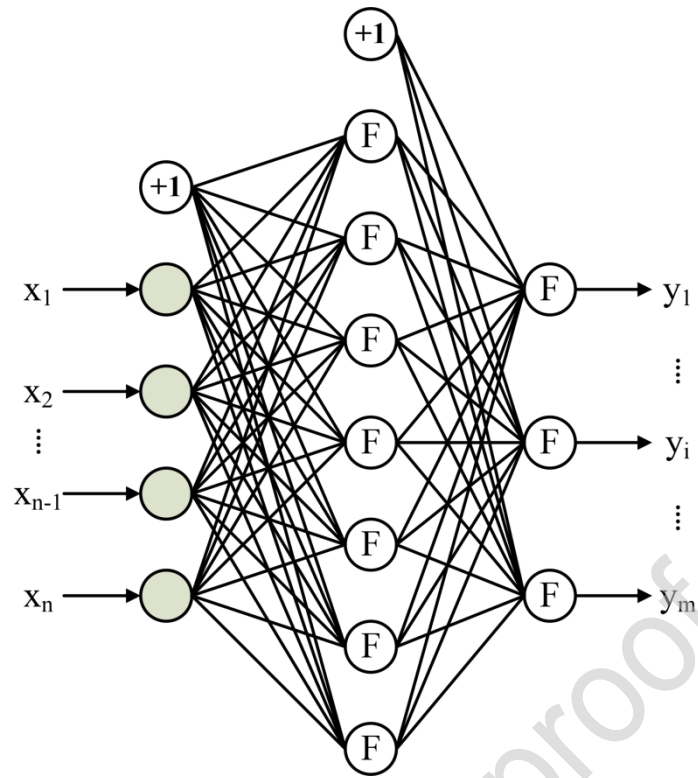


Fig. 4 Schematic of an MLP feedforward network with one hidden layer.

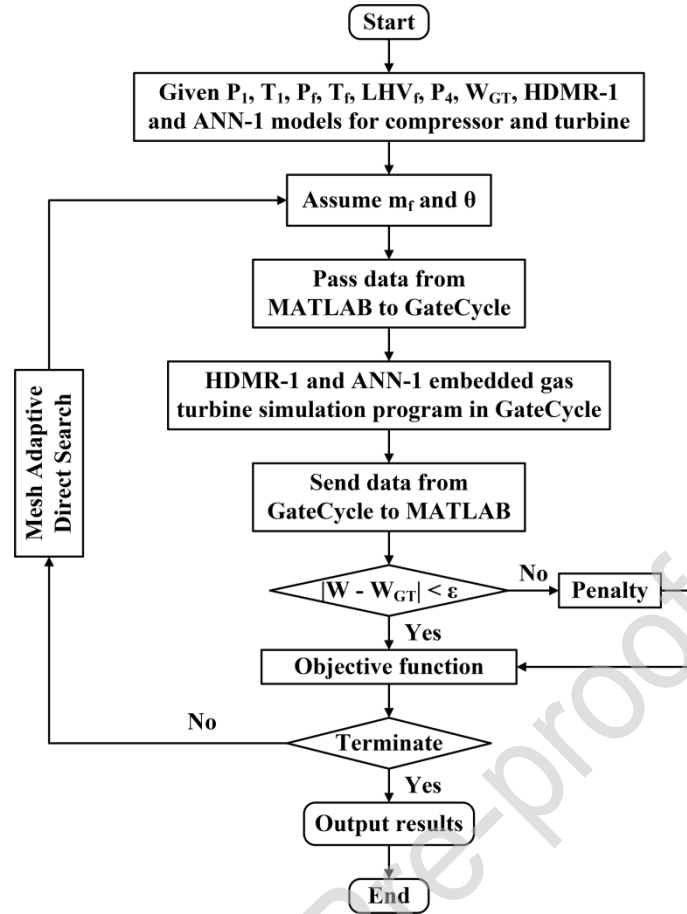


Fig. 5 Performance optimization using GateCycle program augmented with HDMR-1 and ANN-1.

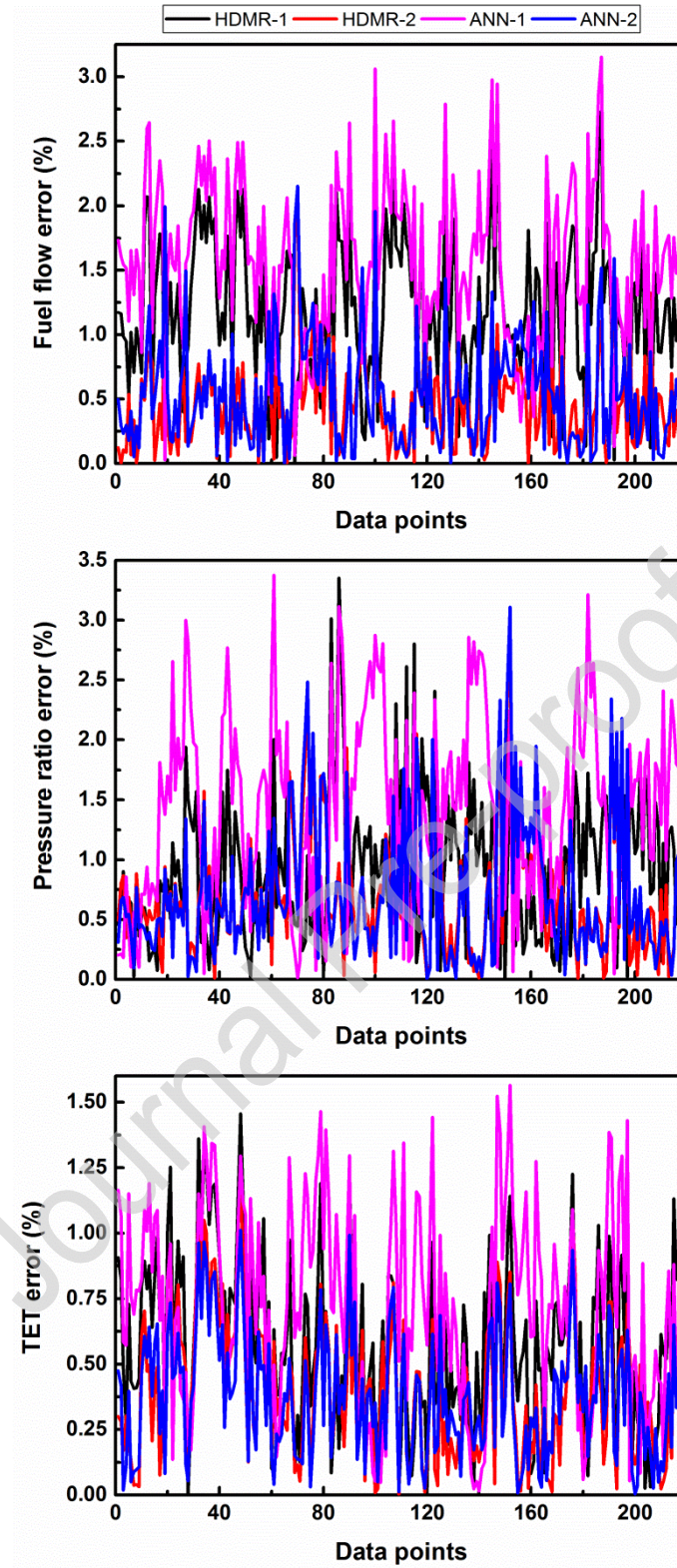


Fig. 6 PR , m_f , and TET relative Errors from reverse evaluation.

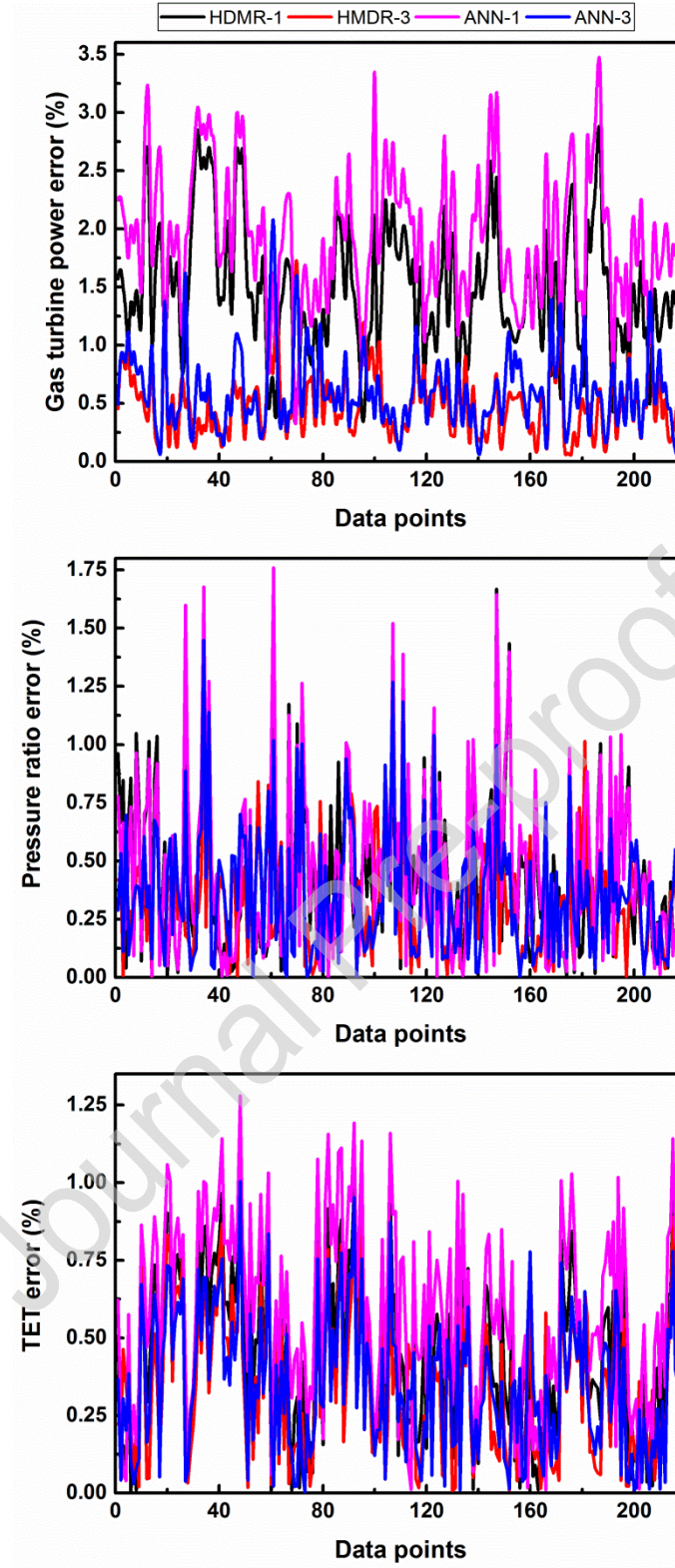


Fig. 7 W_{GT} , PR , and TET relative errors from forward evaluation.

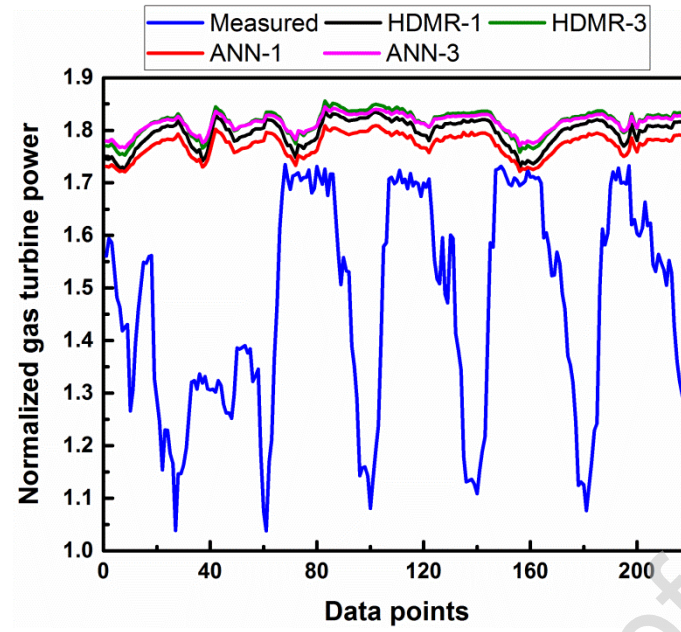


Fig. 8 Measured gas turbine power and predicted power generation capacity.

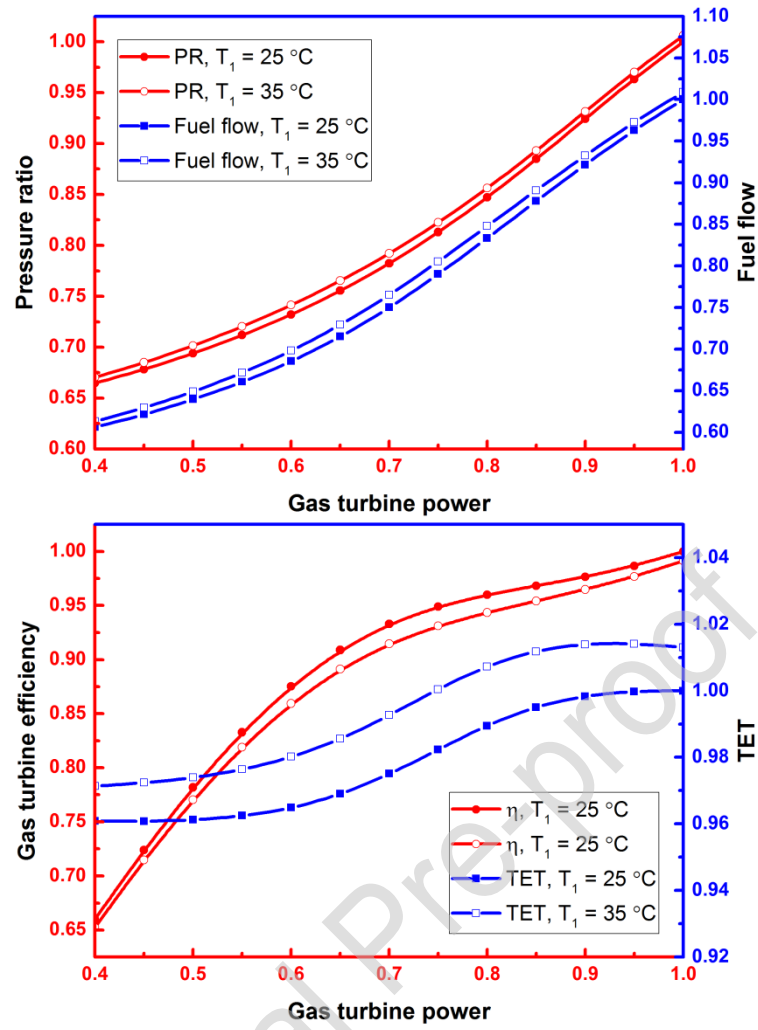


Fig. 9 Normalized GT part-load performance.

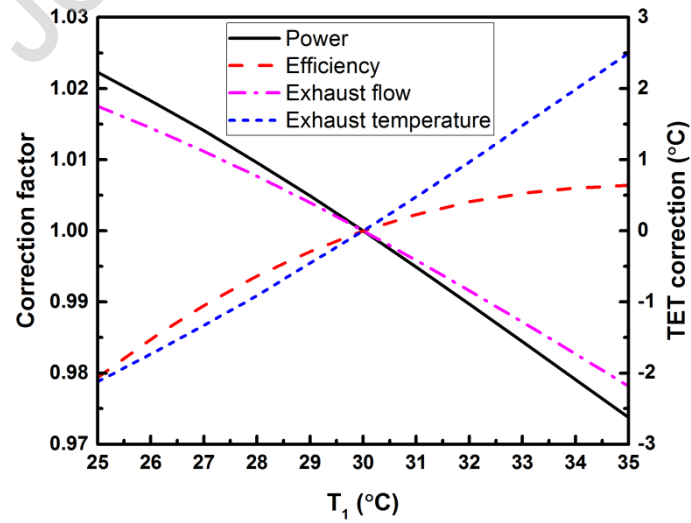


Fig. 10 Corrections of gas turbine parameters on compressor inlet air temperature (T_1).

Highlights

- A data-driven method is developed for predicting gas turbine performance.
- The method builds and uses surrogate models via machine learning from operational data.
- Surrogate models predict gas turbine part-load and full-load performance.
- Gas turbine power generation capacity under any operating condition is obtained.
- Gas turbine performance correction curves are successfully reproduced.

Table 1 Input and output parameters for HDMR-1 and ANN-1 models of compressor and turbine.

	Input parameters	Output parameters
Compressor	x_1 : IGV opening (θ)	$m_{c,cor} = f_1(x_1, x_2, x_3)$
	x_2 : compressor inlet temperature (T_1)	$T_2 = f_2(x_1, x_2, x_3)$
	x_3 : fuel flow (m_f)	
Turbine	x_1 : expansion ratio (π)	$m_{t,cor} = f_3(x_1, x_2)$
	x_2 : turbine inlet temperature (T_3)	$T_4 = f_4(x_1, x_2)$

Table 2 Detailed HDMR-1 models for compressor and turbine.

Coefficients	y	$= C + A_{1,1}x_1 + A_{2,1}x_2 + A_{3,1}x_3 + A_{1,2}x_1^2 + A_{2,2}x_2^2 + A_{3,2}x_3^2 + B_{1,2,1,1}x_1x_2 + B_{1,3,1,1}x_1x_3 + B_{2,3,1,1}x_2x_3$			Eq. (10)
		$m_{c,cor}$	T_2	$m_{t,cor}$	T_4
C		-21335.574	-3034.8	1579.6	388.5
$A_{1,1}$		-66.204	-13.334	0	-49.67
$A_{2,1}$		88.238	12.092	0	0.5188
$A_{3,1}$		3053.843	313.886	0	0
$A_{1,2}$		-0.0256	-1.883×10^{-3}	0	0.7954
$A_{2,2}$		0	0	0	0
$A_{3,2}$		0	0	0	0
$B_{1,2,1,1}$		0.238	0.0434	0	0.01172
$B_{1,3,1,1}$		0.646	0.0487	0	0
$B_{2,3,1,1}$		-8.967	-1.022	0	0
R^2		0.988	0.981	-	0.987

R^2 : Coefficient of determination.

Table 3 Error distribution of HDMR-1 models for compressor and turbine.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
$m_{c,cor}$	2056	567	49	3	0.674	3.389
T_2	2668	7	0	0	0.254	1.172
$m_{t,cor}$	2202	473	0	0	0.594	1.969
T_4	2367	308	0	0	0.751	1.490

Table 4 Detailed HDMR-2 models for predicting GT part-load performance.

Coefficients	$y = C + A_{1,1}x_1 + A_{2,1}x_2 + A_{1,2}x_1^2 + A_{2,2}x_2^2 + B_{1,2,1,1}x_1x_2$			Eq. (10)
	PR	m_f	T_4	m_4
C	11.073	6.547	112.59	658.30
$A_{1,1}$	-1.185×10^{-3}	-4.519×10^{-3}	1.889	-0.429
$A_{2,1}$	-4.277×10^{-2}	-0.0164	2.365	-1.720
$A_{1,2}$	0	0	0	0
$A_{2,2}$	1.867×10^{-4}	8.695×10^{-5}	-3.354×10^{-3}	8.360×10^{-3}
$B_{1,2,1,1}$	8.94×10^{-5}	1.233×10^{-4}	-2.258×10^{-3}	1.684×10^{-3}
R^2	0.990	0.996	0.790	0.980

R^2 : Coefficient of determination. Here, x_1 and x_2 denote T_1 and W_{GT} respectively.

Table 5 Error distribution of HDMR-2 models.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
PR	2044	579	51	1	0.702	3.057
m_f	2287	377	10	1	0.546	3.040
T_4	2395	279	1	0	0.499	2.123
m_4	1668	867	129	11	0.873	3.966

Table 6 Detailed HDMR-3 models for predicting GT full-load performance.

Coefficients	y	$= C + A_{1,1}x_1 + A_{2,1}x_2 + A_{3,1}x_3 + A_{1,2}x_1^2 + A_{2,2}x_2^2 + A_{3,2}x_3^2 + B_{1,2,1,1}x_1x_2 + B_{1,3,1,1}x_1x_3 + B_{2,3,1,1}x_2x_3$					Eq. (10)
		W_{GT}	PR	T_3	T_4	m_4	
C		472.11	-2.284	-941.31	40.85	-1006.42	
$A_{1,1}$		2.248	0.0615	-7.079	-2.348	-3.346	
$A_{2,1}$		-1.649	0.0316	6.433	1.922	4.345	
$A_{3,1}$		-19.29	1.957	181.11	30.19	179.22	
$A_{1,2}$		-1.56×10^{-3}	-8.148×10^{-5}	2.92×10^{-3}	4.13×10^{-3}	-1.375×10^{-3}	
$A_{2,2}$		0	0	0	0	0	
$A_{3,2}$		0	0	0	0	0	
$B_{1,2,1,1}$		-4.24×10^{-3}	-2.715×10^{-6}	0.0273	5.48×10^{-3}	0.012	
$B_{1,3,1,1}$		-4.76×10^{-2}	1.780×10^{-3}	-0.154	-0.0372	0.0376	
$B_{2,3,1,1}$		0.118	-4.966×10^{-3}	-0.460	-0.0230	-0.522	
R^2		0.997	0.998	0.975	0.861	0.989	

R^2 : Coefficient of determination. Here, x_1 , x_2 and x_3 denote θ , T_1 and m_f respectively.

Table 7 Error distribution of HDMR-3 models.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
W_{GT}	2158	492	25	0	0.614	2.988
PR	2596	79	0	0	0.352	1.681
T_3	2620	55	0	0	0.315	1.538
T_4	2527	148	0	0	0.398	1.920
m_4	2108	525	39	3	0.647	3.247

Table 8 Training parameters of ANN models.

Parameter	Value
Neural network	MLP
Hidden layer number	1
Hidden layer neuron	2-20 with a step of 2
Activation function	hidden / output layer: tansig / purelin
Loss function	MSE
Training algorithm	traingdx
Training data division	Training / validation : 80% / 20%
Maximum epochs	30000
Regularization ratio	0.01
Maximum validation checks	300

Table 9 Error distribution of ANN-1 models for compressor and turbine.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
$m_{c,cor}$	2021	600	50	4	0.689	3.649
T_2	2674	1	0	0	0.265	1.066
$m_{t,cor}$	2202	473	0	0	0.594	1.969
T_4	2675	0	0	0	0.123	0.666

Table 10 Error distribution of ANN-2 model.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
PR	1937	667	70	1	0.736	3.108
m_f	2168	479	24	4	0.601	4.349
T_4	2416	258	1	0	0.481	2.031
m_4	1653	873	136	13	0.882	4.105

Table 11 Error distribution of ANN-3 model.

Error (%)	≤ 1	1 - 2	2 - 3	> 3	Average	Maximum
W_{GT}	2032	590	51	2	0.690	3.799
PR	2568	107	0	0	0.368	1.760
T_3	2616	59	0	0	0.313	1.576
T_4	2516	159	0	0	0.396	1.757
m_4	2083	536	52	4	0.667	3.506

Table 12 Comparison of HDMR and ANN models.

Model	Number of parameters	Sum of squared errors	AIC
HDMR-1	44	1.338×10^6	1.982×10^3
HDMR-2	24	1.956×10^4	1.025×10^3
HDMR-3	95	1.785×10^4	1.147×10^3
ANN-1	63	2.187×10^6	2.126×10^3
ANN-2	46	1.152×10^4	9.538×10^2
ANN-3	77	1.729×10^4	1.104×10^3

Table 13 Gas turbine reference parameters and performance.

Parameter	Notation	Value
Air pressure (kPa)	P_a	101.32
Air relative humidity (%)	RH	85
Inlet temperature (°C)	T_1	30
Fuel flow (kg s ⁻¹)	m_f	14.82
Fuel lower heating value (kJ kg ⁻¹)	LHV_f	46842
Power (MW)	W_{GT}	225.6
Efficiency (%)	η_{GT}	32.50
Exhaust flow (kg s ⁻¹)	TEF	680.0
Exhaust temperature (°C)	TET	604.2